

ORIGINAL ARTICLE

Individual differences in decisiveness: ERP correlates of response inhibition and error monitoring

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Email: magdalena.senderecka@uj.edu.pl**Funding information**

Polish Ministry of Science and Higher Education Iuventus Plus grant (0486/IP3/2011/71) (data collection), National Science Centre of Poland Opus 10 grant (2015/19/B/HS6/00341) (to M. S.), National Science Centre of Poland Maestro grant (DEC 2011/02/A/HS6/00155) (to M. K.)

Abstract

The aim of the current study was to examine whether and how self-reported decisiveness is associated with response inhibition and performance monitoring. We hypothesized that these two cognitive control mechanisms, both of which are often associated with decision making, would differ in individuals varying in decisiveness. We focused on ERP correlates and behavioral measures of inhibition and error processing in the stop-signal task. We expected a negative relationship between decisiveness and behavioral measures of inhibitory control. We also hypothesized that stop-signal-locked N1 and P3 components and response-locked error-related negativity (ERN) would be less pronounced when participants self-reported higher levels of decisiveness. Correlation analysis identified an association between high decisiveness, long stop-signal reaction time, and low inhibition rate. Analysis with mixed-effects linear models revealed that stop signals evoked less pronounced N1 and P3 in individuals scoring higher on decisiveness in both successfully and unsuccessfully inhibited trials. Additionally, high decisiveness was linked to reduced error monitoring, as indicated by decreased ERNs. Importantly, we also found positive association between P3 onset latency and decisiveness, suggesting that individuals scoring higher on this measure have relatively less ability to rapidly engage the stopping process. Thus, our findings primarily indicate that decisiveness is negatively associated with the efficiency of both response inhibition and error monitoring. They also suggest that highly decisive people may share some characteristics of diminished cognitive control with impulsive individuals.

KEYWORDS

decisiveness, ERPs, error monitoring, response inhibition, stop-signal task

1 | INTRODUCTION

The ability to make decisions quickly and effectively is a valuable skill in a rapidly changing and uncertain environment. Individuals differ in how quickly they make decisions: some hesitate or deliberately postpone a decision, while others can act on the spot in a more impulsive way. This tendency is characterized as decisiveness and is usually measured using questionnaires (e.g., Webster & Kruglanski, 1994). In the present study, we aimed to evaluate whether and how self-reported decisiveness is associated with inhibitory processing and performance monitoring.

These two cognitive functions are likely to be associated with decisiveness for several reasons. Effective decision making requires finding a balance between competing needs for urgency and caution (Evans, Rae, Bushmakin, Rubin, & Brown, 2017). Making more decisions per time unit may maximize the probability of obtaining a reward. In this sense, quick decisions can maximize reward very efficiently; however, they are also frequently wrong, particularly when they arise from haste or thoughtlessness, which points to a lack of inhibitory control. On the contrary, slow decisions can lead to fleeting opportunities being missed, but they can also allow negative consequences of impulsive errors to be

avoided. Thus, decision making is inevitably associated with both abilities: to inhibit or avoid certain behaviors or thoughts (Botvinick, 2007; Ridderinkhof, Van Den Wildenberg, Segalowitz, & Carter, 2004) and to detect and evaluate an error in order to take remedial action and influence future decisions (Nieuwenhuis, Holroyd, Mol, & Coles, 2004; Ridderinkhof, Van Den Wildenberg et al., 2004).

Previous studies suggest that the tendency to make rapid decisions is a consistent correlate of a variety of impulsive behaviors, which in turn have been associated with reduced efficiency of response inhibition and performance monitoring mechanisms (e.g., Bekker, Overtom et al. 2005; Liotti, Pliszka, Perez, Kothmann, & Woldorff, 2005; Overtom et al., 2002). To our knowledge, no study has examined these two crucial cognitive control mechanisms in relation to individual differences in decisiveness. Moreover, despite the growing interest in the cerebral basis of decision making (for reviews, see Krawczyk, 2002; Venkatraman & Huettel, 2012; Wichary & Smoleń, 2016), no study has looked at decisiveness-related neural correlates of response inhibition and error monitoring. The aim of this study is to fill this gap by investigating the links between decisiveness and selected behavioral measures, as well as ERP components, which represent well-validated indexes of cognitive control functions.

Decisiveness was proposed by Webster and Kruglanski (1994) as one of the five facets of their need for cognitive closure (NFC) construct. In this framework, decisiveness—measured on a self-report subscale—is understood as the urgency to formulate judgments and make decisions quickly and confidently, regardless of their importance. This urgency is motivated by the desire to reach cognitive closure, that is, to have “an answer on a given topic, any answer (...) compared to confusion and ambiguity” (Kruglanski, 1990, p. 337). Although originally conceived as a facet of NFC, decisiveness has also been studied in its own right due to its weak association with other facets of NFC (e.g., Neuberg, Judice, & West, 1997; Roets & Van Hiel, 2007). Moreover, in other research traditions, decisiveness has been measured with various scales (usually providing a score of indecisiveness) that focus on general decision-making tendencies (e.g., Frost & Shows, 1993; Germeijs & De Boeck, 2002).

Results of questionnaire studies suggest that trait decisiveness is related to low anxiety, high self-confidence, and low neuroticism (Germeijs & Verschueren, 2011; Germeijs, Verschueren, & Soenens, 2006). It has been observed that participants high in decisiveness are less likely to interpret ambiguous situations as threatening, thus indicating that they presumably are, in general, less worried than average (Rassin & Muris, 2005). In addition, experimental research indicates that decisive individuals take less time to make decisions (Ferrari & Pychyl, 2007; Frost & Shows, 1993). Using a multiattribute choice task, in our previous work we also showed that decisive individuals gather less information

before making a choice and rely on a simpler decision heuristic than indecisive people, thus indicating their decreased need for information and tendency to simplify the decision process (Wichary, Kossowska, Orzechowski, Ślifierz, & Marković, 2008). Previous studies also suggest that quick decision making, as measured behaviorally by faster and less accurate responses during cognitive tasks, is consistently associated with a higher level of trait impulsivity (Dickman, 1990; Edman, Schalling, & Levander, 1983; Gerbing, Ahadi, & Patton, 1987). The positive correlation between these two measures indicates that both decisiveness and impulsivity may represent a maladaptive extreme of decision making underpinned by differences in cognitive control.

However, it should be noted that the relation between decisiveness and impulsivity is not always as straightforward as is implied by the previously reviewed studies. For instance, Bar-Tal and Kossowska (2010) found a positive correlation between decisiveness and the psychometric measure of functional impulsivity, which is defined as the tendency to act with relatively little forethought when this seems to be optimal. However, they did not observe any significant correlation between decisiveness and dysfunctional impulsivity (i.e., the urgency to behave impulsively even in situations that require deliberation). In the same vein, Barkley-Levenson and Fox (2016) found mixed results regarding the relationship between the NFC subscale of decisiveness and different types of impulsiveness. The results ranged from a positive correlation of decisiveness with functional impulsivity, through a nonexistent relationship with sensation seeking, to a negative association with dysfunctional impulsivity. Altogether, these studies make a valuable contribution by showing that decisiveness and impulsivity are closely intertwined; however, there are many possible directions that this association can take. These discrepancies point to the strong need for research exploring the direct link between cognitive control mechanisms and decisiveness itself.

As already mentioned, a substantial body of evidence shows that poor impulse control correlates significantly with less effective response inhibition and decreased error monitoring (e.g., Bekker, Overtom et al. 2005; Liotti et al., 2005; Overtom et al., 2002). The efficiency of these two cognitive control functions is frequently investigated using the stop-signal task (SST), which requires participants to make two-choice reactions to “go” stimuli but to suppress a prepotent motor response on the infrequent presentation of a “stop” signal (Logan, 1994). Previous SST studies have identified longer stop-signal reaction times (SSRT) in both adults and children with low impulse control who were diagnosed with attention deficit hyperactivity disorder (for a meta-analytic review, see Lijffijt, Kenemans, Verbaten, & van Engeland, 2005). However, mean reaction times for the correct go trials (go RT) were frequently found to be neither prolonged nor shortened in clinically impulsive adults and

children. Previous SST studies have also reported significantly reduced stop-signal and erroneous-response-related ERPs in these groups, that is, less pronounced N1, P3, and error-related negativity (ERN; e.g., Bekker, Overtom et al. 2005; Liotti et al., 2005; Overtom et al., 2002; Senderecka, Grabowska, Szewczyk, Gerc, & Chmylak, 2012). Most of these findings were confirmed in research on healthy impulsive individuals using the SST or go/no-go tasks (Heritage & Benning, 2013; Logan, Schachar, & Tannock, 1997; Marsh, Dougherty, Mathias, Moeller, & Hicks, 2002; Ruchow, Spitzer, Grön, Grothe, & Kiefer, 2005; Ruchow et al., 2008; Shen, Lee, & Chen, 2014; Stahl & Gibbons, 2007; however, note that some studies yielded the opposite results, Dimoska & Johnstone, 2007; Lansbergen, Böcker, Bekker, & Kenemans, 2007).

In the present study using the SST task, we aimed to evaluate whether and how inhibitory processing and performance monitoring are associated with self-reported decisiveness. Given that the urgency to make decisions quickly is correlated with trait impulsivity (Dickman, 1990; Edman et al., 1983; Gerbing et al., 1987), which in turn is usually associated with reduced efficiency of response inhibition and performance monitoring, we hypothesized a negative correlation between the indices of these two cognitive control mechanisms and decisiveness.

To assess auditory stop-signal-related brain activity, we used two ERP components previously studied with the SST, namely, N1 and the P3. The auditory N1 is a sharp negative deflection that peaks over the frontocentral or central regions and that usually begins at 60–80 ms and lasts until 160 ms after the onset of the stop sound (Näätänen & Picton, 1987; Woods, 1995). It originates mainly in the auditory cortex; it reflects the initial stages of sensory analysis of the stimulus and is sensitive to selective attention (Hillyard, Hink, Schwent, & Picton, 1973). In the SST context, it has also been considered a marker of stop-signal-driven shift of attention, which partly determines the subsequent success of response inhibition (Bekker, Kenemans, Hoeksma, Talsma, & Verbaten, 2005). The P3 component, which in the SST peaks at around 300–350 ms, resembles the classical P3b with a more central distribution (Kok, Ramautar, De Ruiter, Band, & Ridderinkhof, 2004). It has been implicated as an index of inhibitory mechanisms (e.g., Bekker, Kenemans et al., 2005; De Jong, Coles, Logan, & Gratton, 1990).

To examine error-related brain activity, we used the ERN; this is a sharp negative wave that peaks at around 0–100 ms after the onset of an erroneous reaction and is distributed over frontocentral regions (Falkenstein, Hohnsbein, Hoormann, & Blanke, 1990; Gehring, Goss, Coles, Meyer, & Donchin, 1993). Various models implicate the ERN as a reflection of the mechanism that monitors the difference between the intended and the actually performed action (Coles, Scheffers, & Holroyd, 2001; Falkenstein, Hohnsbein,

Hoormann, & Blanke, 1991; Gehring et al., 1993), as an index of conflict between simultaneously active correct and incorrect response tendencies (Botvinick, Braver, Barch, Carter, & Cohen, 2001; Carter & van Veen, 2007; Yeung, Botvinick, & Cohen, 2004), or as a signal of reinforcement learning (Holroyd & Coles, 2002). In addition, the ERN has also been proposed to reflect the subjective significance or aversiveness of an error (Hajcak & Foti, 2008; Hajcak, Moser, Yeung, & Simons, 2005).

To explore correct-response-related brain activity, we used the CRN (correct-related negativity), which occurs in the same time range as the ERN (i.e., shortly after correct response execution) and peaks at frontocentral electrode sites (Falkenstein, Hoormann, Christ, & Hohnsbein, 2000; Ford, 1999; Vidal, Burle, Bonnet, Grapperon, & Hasbroucq, 2003; Vidal, Hasbroucq, Grapperon, & Bonnet, 2000). This negative wave has been considered to reflect higher uncertainty about the correctness of the actual response (Pailing & Segalowitz, 2004a; Scheffers & Coles, 2000), enhanced response-checking tendency (Falkenstein et al., 2000; Vidal et al., 2000), or postresponse implementation of cognitive control (Bartholow et al., 2005; Bonnefond, Doignon-Camus, Hoeft, & Dufour, 2011; Tillman & Wiens, 2011).

We expected that stop-signal-locked N1 and P3 and response-locked ERN would be less pronounced when participants self-reported a higher level of decisiveness. We did not have any specific hypothesis about the relationship between decisiveness and CRN. In addition, our hypothesis stated that decisiveness would correlate with behavioral performance, pointing to an association between increased SSRT, decreased inhibition rate, and the self-reported tendency to make decisions hastily. Based on previous research (Lijffijt et al., 2005), no significant correlation was expected between go RT and decisiveness.

2 | METHOD

2.1 | Participants

Sixty-five young volunteers (43 female, 22 male) aged 19–33 years old ($M = 21.7$ years, $SD = 2.6$) participated in the study. All participants were in good health, free of medications, and had normal or corrected-to-normal vision. None reported a history of psychiatric or neurological diseases. Of the initial sample recruited for the study, eight participants were excluded from the analyses due to probable misunderstanding of the instructions leading to a lack or low number (six or less) of correctly inhibited responses; one participant was excluded because his mean RT deviated substantially from the mean of the sample (more than 3.0 standard deviations); another one was excluded because of technical problems with the EEG recording. The remaining 55 participants

(38 female, 17 male) were aged 19–27 years old with a mean age of 21.5 years ($SD = 2.1$).

2.2 | Procedure and task

The experimental procedure was in accordance with the ethical principles of the 1964 Declaration of Helsinki (World Medical Organization, 1996). Participants were seated in a dimly lit, sound-attenuated, air-conditioned testing room. After providing written informed consent to participate in the study, they completed the shortened Polish version of the NFC scale (Webster & Kruglanski, 1994; Polish adaptation, Kossowska, 2003; Kossowska, Hanusz, & Trejtowicz, 2012). The scale includes five subscales: preference for order, preference for predictability, discomfort with ambiguity, closed mindedness, and decisiveness. It consists of 15 statements to which participants responded on a 6-point scale from 1 = *strongly disagree* to 6 = *strongly agree*. The decisiveness subscale from the shortened Polish version of the NFC (Kossowska et al., 2012) has acceptable reliability (internal consistency: Cronbach's $\alpha = .65$; mean variance explained using confirmatory factor analysis was 41%). Scores on the full and shortened version of the decisiveness subscale correlated highly with each other ($r = .92$) in a large community sample ($N = 1,754$). The criterion validity test revealed that both decisiveness subscales are similarly associated with specific personality characteristics. Moreover, the scale was adapted to the cultural context in which the study was conducted. The internal consistency of the decisiveness subscale that was obtained on the current sample was good (Cronbach's $\alpha = .78$, $M = 3.3$, $SD = 1.2$). We focused on this subscale in all our analyses.

Participants completed the stop-signal task with auditory stop stimulus. The task required them to perform a primary binary-choice response task. It included a visual go stimulus consisting of a small green square presented with the inner edge at two degrees either to the right or the left of the fixation point. The green square was presented randomly on the left or right side of a 23" computer monitor with 50% probability, for 100 ms, on a black background, approximately 80 cm in front of the participant, at eye level. Participants were instructed to indicate the location of the square by pressing the left or right *Ctrl* key using their index fingers. In addition, they were instructed to react to the go stimulus as quickly and as accurately as possible. Each trial began with the presentation of a white central fixation cross for 800 ms, immediately followed by the presentation of the square.

In a random sample of 25% of the trials, a 1400 Hz tone, which served as the stop signal, was presented binaurally over Sennheiser HD 429 headphones (intensity 60 dB SPL, duration 100 ms, rise and fall time 10 ms). The sound prompted the participants to inhibit their responses to the primary go task, regardless of the location of the square. The

interval between the presentation of the go stimulus and the stop signal (i.e., the stop-signal delay, SSD) was varied trial by trial using a tracking method: the interval increased or decreased by 50 ms (from 100 to 400 ms) for the next stop-signal trial, depending on whether the participants successfully or unsuccessfully inhibited their response to the go stimulus. Thus, there were seven possible SSDs: 100, 150, 200, 250, 300, 350, 400 ms. After a successful inhibition, the interstimulus interval became longer; after an unsuccessful inhibition, it became shorter. The initial value of the SSD was set to 150 ms. The aim of the tracking method was to converge on an SSD at which participants successfully inhibited responses to approximately 50% of the stop-signal trials. Figure 1 presents an outline of the stop-signal task design.

Participants received one or two practice blocks of 24 trials before data collection to ensure that they understood the task. After the practice runs, they completed four experimental blocks, each consisting of 50 trials, with short breaks in between. All participants were asked to restrict body movements and blinking as much as possible during the recording of the EEG. The task was implemented using DMDX software (Forster & Forster, 2003).

2.3 | EEG recording

The continuous scalp EEG was recorded from 32 silver/silver-chloride (Ag/AgCl) active electrodes (with preamplifiers) using the BioSemi ActiveTwo system. The electrodes were secured in an elastic cap (Electro Cap) according to the extended 10–20 international electrode placement system. The zero-reference principal voltage values (each site quantified relative to the driven right leg and common mode sense loop) were digitized at a sampling rate of 256 Hz. The horizontal and vertical electrooculograms (EOGs) were monitored using four additional electrodes placed above and below the right eye and in the external canthi of both eyes. All channels were rereferenced offline to the average of the two mastoid electrodes. The recordings were filtered offline with a high-pass filter of 0.05 Hz (slope 24 dB/oct) and a low-pass filter of 25 Hz (slope 12 dB/oct). Ocular and other stationary artifacts were removed with the independent component analysis algorithm (ICA) using the BrainVision Analyzer 2 (Brain Products, Munich, Germany).

2.4 | Data quantification

Stimulus-locked (–100 ms to 700 ms around the stop-signal onset) and response-locked (–150 ms to 600 ms relative to the key press) segments were subsequently checked and averaged. Contaminated trials exceeding maximum/minimum amplitudes of $\pm 65 \mu V$ were rejected by a semiautomatic procedure. Stop-signal ERPs were aligned to the prestimulus baseline from –100 ms to 0 ms, whereas motor

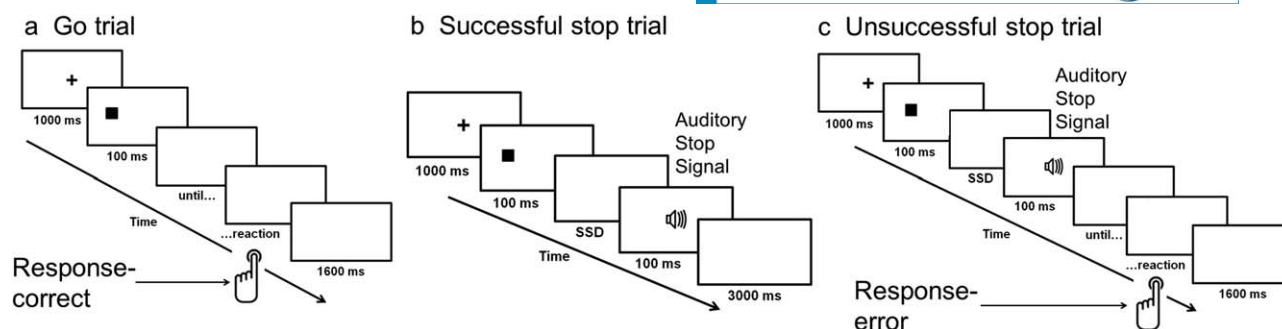


FIGURE 1 Behavioral task. (a) Go trial without stop-signal presentation. (b) Successfully inhibited stop-signal trial. (c) Unsuccessfully inhibited stop-signal trial. SSD = stop-signal delay

reaction ERPs were baseline-corrected relative to the pre-response interval from -150 ms to -50 ms. Stop-signal ERPs were averaged separately for each type of stop-signal trial: successful stop and unsuccessful stop. Motor reaction ERPs were calculated separately for correct responses to go stimuli (response-correct) and unsuccessfully inhibited responses in failed stop trials (response-error). The mean number of correct, artifact-free epochs included in the ERP analysis across all participants for each of the stop-signal and response trial categories was as follows: successful stop, $M = 18.0$ ($SD = 6.3$); unsuccessful stop, $M = 29.1$ ($SD = 7.0$); response-correct, $M = 141.3$ ($SD = 11.6$); response-error, $M = 29.1$ ($SD = 7.0$). The minimum number of epochs was 7 for successful stop, 15 for unsuccessful stop, 83 for response-correct, 15 for response-error. Participants with a low number of valid trials in one condition were included in our mixed models (see below for more details) because this kind of analysis works well for unbalanced data sets. In the SST, the ERPs elicited in response to the go and stop stimuli overlap in time due to the short interval between their presentation. To control the possible distortion resulting from the overlap across two stop-signal conditions, we included a 7-level categorical stop delay predictor into regression analyses (see below).

Consistent with our previous research using the SST paradigm (Senderecka, 2018), time windows were selected around the N1 (120–190 ms) and the P3 (270–400 ms), both of which were time-locked to the stop-signal presentation, and the ERN/CRN (0–80 ms), which was time-locked to the motor reaction. Mean voltage amplitudes in the component-specific windows were used for statistical analysis. For each of these components, appropriate electrode clusters were selected based on the literature (see Näätänen & Picton, 1987, for the auditory N1 literature review; Kok et al., 2004, for the inhibitory P3; Overbeek, Nieuwenhuis, & Ridderinkhof, 2005, for the ERN/CRN). The N1 component was analyzed at the averaged central sites (FC1, FC2, C3, C4, Cz), the P3 was analyzed at the centroparietal sites (Cz, CP1, CP2, Pz), and the ERN/CRN was analyzed at the averaged frontocentral sites (Fz, FC1, FC2, Cz).

2.5 | Statistical analyses

SSRT, which provides an estimate of the latency of the inhibitory process (Verbruggen & Logan, 2008), was calculated following the procedure of Logan (1994). Reaction times from go stimuli responses on which no stop signal occurred were collapsed into a single distribution and rank ordered. The n th reaction time was selected, where n was obtained by multiplying the number of no-signal reaction times in the distribution (150) by the probability of responding (e.g., 0.5 if the global inhibition rate was equal to 50%) for each participant separately. SSRT was calculated by subtracting the average SSD from the n th reaction time, following the horse race model (see Logan & Cowan, 1984; Verbruggen & Logan, 2008, for more detail).

The distribution of decisiveness scores and behavioral variables examined with the Kolmogorov–Smirnov test was not statistically different from the normal distribution. To test go and stop task performance in participants varying with respect to decisiveness, we ran a series of Pearson correlation analyses between decisiveness and the behavioral variables: go RT, inhibition rate, SSRT, and SSD. To analyze the amplitudes of the N1, P3, and ERN/CRN components, we built a series of linear mixed effects (LME) models with participants as the random variable. Using LME models allowed us to simultaneously measure subject-related variable decisiveness as well as to control for trial-related effects such as stop delay (accounting for potential overlap with the ERPs to go stimuli) or trial number (accounting for effects of training, fatigue, or developing a task strategy). Moreover, the linear mixed effects account for shrinkage toward the mean and work well with unbalanced data (e.g., for participants with an unbalanced number of successfully and unsuccessfully inhibited responses in stop trials).

Each LME model included decisiveness and trial number as continuous fixed effects. Moreover, the models for the N1 and P3 components included as fixed effects trial type (successful stop vs. unsuccessful stop), the interaction between trial type and decisiveness, the interaction between trial type

and trial number, and the categorical effect stop delay. The latter variable was treated as a regressor of no interest. The model for the ERN/CRN included response type (response-correct vs. response-error), the interaction between response type and decisiveness, and the interaction between response type and trial number.

All LME models had maximal random effect structure, but with all correlations of random effects removed.¹ All categorical predictors were deviation coded (and converted to dummy variables when necessary); the continuous predictors were centered prior to entering the analyses. Absolute *t* values greater than the conventional level of 2 were considered significant. All analyses were conducted using the lme4 package for R (version 1.1–15; Bates, Mächler, Bolker, & Walker, 2015). All models converged without any need for trimming; the final models had random effects with zero variance removed.

3 | RESULTS

3.1 | Behavioral data

The mean reaction time of the correct go trials was 374.0 ms (*SD* = 64.1). The inhibition rate was 38.0% (*SD* = 13.5), indicating that stop performance was slightly lower than expected. SSD was 161.1 ms (*SD* = 48.0), whereas SSRT was 231.7 ms (*SD* = 50.4). Correlation analyses revealed that decisiveness was positively correlated with SSRT ($r = .38, p = .004$) and negatively correlated with inhibition rate ($r = -.38, p = .004$), indicating lower inhibitory performance in highly decisive participants. We did not observe a significant correlation between questionnaire data and two other behavioral variables: go RT ($r = -.09, p = ns$) and SSD ($r = -.19, p = ns$).

3.2 | ERP time-locked to stop-signal presentation

Figure 2 shows the grand-averaged ERPs time-locked to the stop-signal presentation at representative sites with scalp distribution maps for difference waves. The grand-averaged ERPs were obtained from averaging single trials.

¹For example, the lmer formula used to analyze the amplitude of the N1 component was the following: $N1 \sim \text{TrialNumber} + \text{TrialType} + \text{Decisiveness} + \text{TrialNumber}:\text{TrialType} + \text{Decisiveness}:\text{TrialType} + \text{StopDelay100.150} + \text{StopDelay100.200} + \text{StopDelay100.250} + \text{StopDelay100.300} + \text{StopDelay100.350} + \text{StopDelay100.400} + (1|\text{Subject}) + (0 + \text{TrialType}|\text{Subject}) + (0 + \text{TrialNumber}|\text{Subject}) + (0 + \text{TrialType}:\text{TrialNumber}|\text{Subject}) + (0 + \text{StopDelay100.150}|\text{Subject}) + (0 + \text{StopDelay100.200}|\text{Subject}) + (0 + \text{StopDelay100.250}|\text{Subject}) + (0 + \text{StopDelay100.300}|\text{Subject}) + (0 + \text{StopDelay100.350}|\text{Subject}) + (0 + \text{StopDelay100.400}|\text{Subject})$. The formulas for the remaining analyses were altered as described in the text.

3.2.1 | Amplitude of the N1 component (120–190 ms)

The results of the mixed-effects analysis of the mean amplitude in the N1 time window are shown in Table 1. The analysis revealed a main fixed effect of trial type. We found a larger N1 amplitude in successful stop trials than in unsuccessful stop trials. We also observed a main fixed effect of decisiveness: the negative deflection was less pronounced in both successfully and unsuccessfully inhibited trials when participants scored higher on the decisiveness scale. Trial Type $\times$ Decisiveness interaction effect did not reach significance. Figure 3 visualizes the predictions of the model for the interaction of decisiveness and trial type.

3.2.2 | Amplitude of the P3 component (270–400 ms)

The LME model for the mean amplitude of the P3 component is shown in Table 2. The global analysis conducted for P3 amplitude revealed a significant main effect of trial type. We observed a larger P3 in successful stop trials than in unsuccessful stop trials. We also found a main effect of decisiveness: positive deflection was less pronounced in participants higher in decisiveness relative to participants lower in decisiveness. The interaction between decisiveness and trial type, as well as the interaction including trial number, were not significant. The model estimates for trial type and decisiveness interaction are shown in Figure 4.

3.3 | ERP time-locked to motor response

Figure 5 presents the grand-averaged ERPs time-locked to the motor response at representative sites with scalp distribution maps for difference waves. The grand-averaged ERPs were obtained from averaging single trials.

3.3.1 | Amplitude of the ERN/CRN component (0–80 ms)

The results of the mixed-effects analysis of the mean amplitude in the ERN/CRN time window are shown in Table 3. We observed that the ERPs to response-error showed a sharp negative peak that was less pronounced in the ERPs to response-correct. The frontocentral postresponse negativity also showed a main effect of decisiveness: the negative deflection in the ERN/CRN time window was reduced in participants scoring higher on decisiveness, relative to individuals scoring lower on decisiveness. We also found a Response Type $\times$ Decisiveness interaction effect, indicating that, with increasing decisiveness, the difference between the ERN and CRN decreased. The slope of CRN amplitudes depending on decisiveness ($1.34 - 0.5 * 0.98 = 0.85$) was

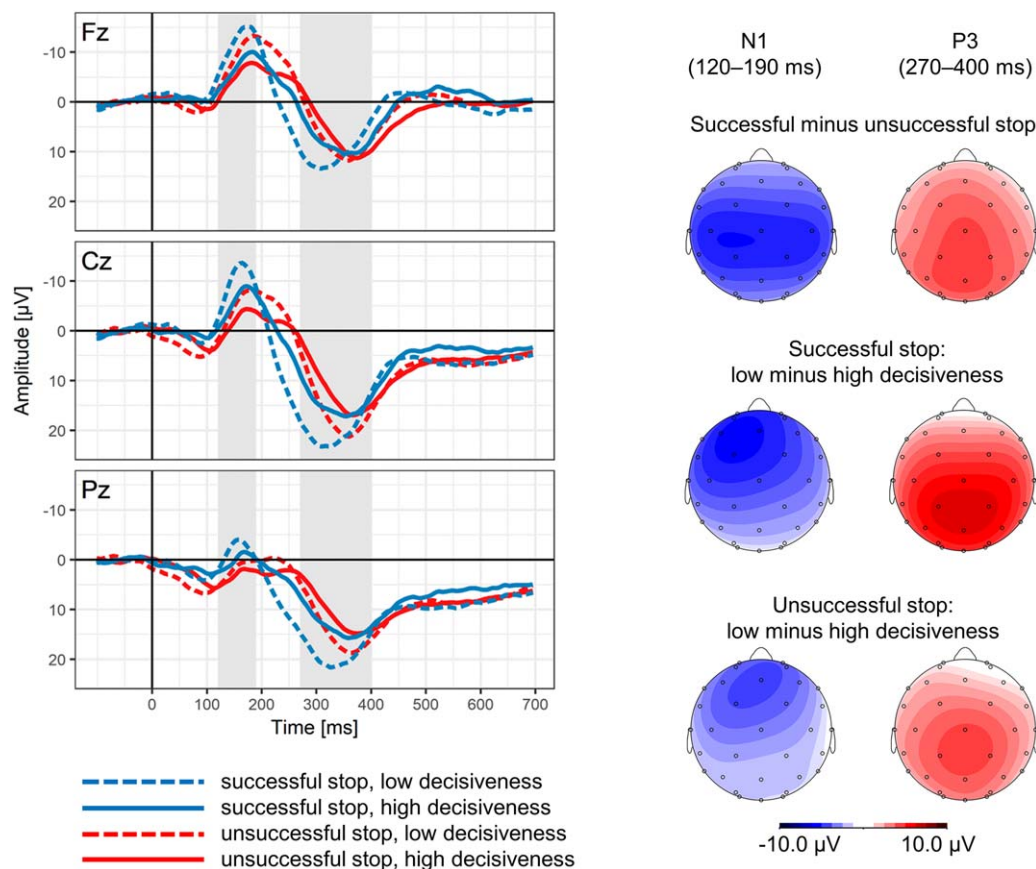


FIGURE 2 Stop-signal-locked grand-averaged waveforms from all trials at representative midline electrodes Fz, Cz, and Pz (left) with scalp potential difference maps for the N1 and P3 components (right). Two groups of participants scoring high and low on decisiveness were formed by a median split procedure only for illustrative purposes. The component-specific windows examined in this study are highlighted

TABLE 1 Estimates of the LME model of the N1 amplitude

Fixed effects				Random effects (by subject)
Effect	Estimate	SE	<i>t</i> value	SD
Intercept	−7.63	0.77	−9.96	4.43
Trial number	−0.27	0.17	−1.61	0.52
Trial type	4.30	0.51	8.48	1.81
Decisiveness	1.29	0.57	2.28	
Stop delay 100 vs. 150	3.43	0.59	5.79	1.73
Stop delay 100 vs. 200	2.64	0.83	3.17	2.96
Stop delay 100 vs. 250	−1.07	1.08	−0.99	3.49
Stop delay 100 vs. 300	−2.93	1.12	−2.61	0.77
Stop delay 100 vs. 350	−0.77	2.42	−0.32	8.48
Stop delay 100 vs. 400	−5.42	1.90	−2.86	
Trial number:Trial type	−0.05	0.30	−0.18	
Trial type:Decisiveness	−0.59	0.44	−1.36	
Residual				10.26

Note. Significant effects are indicated in bold.

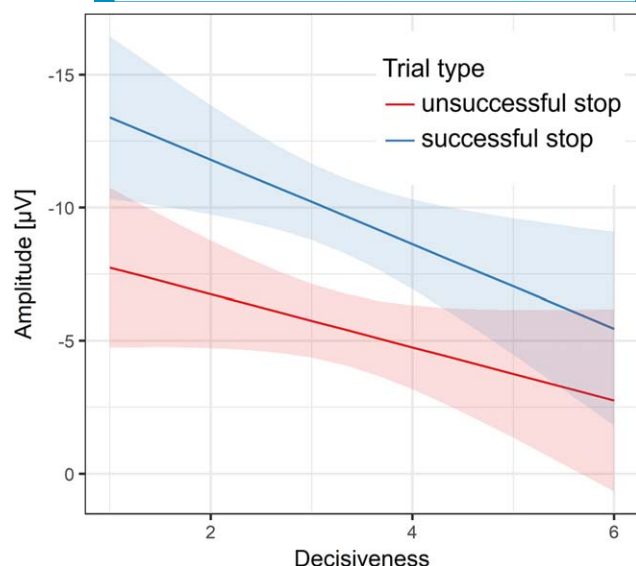


FIGURE 3 Predictions of the LME estimates of the N1 amplitude. Colored bands represent 95% confidence intervals based on the model. Values on the y axis are inverted to correspond with ERP plots

more attenuated than the slope of ERN amplitudes ($1.34 + 0.5 * 0.98 = 1.83$). In other words, every point on the decisiveness subscale was accompanied by an increase of $0.85 \mu\text{V}$ in the CRN and of $1.83 \mu\text{V}$ in the ERN. Direct tests conducted separately for the response-error and response-correct conditions revealed that the slope of ERN amplitudes in the function of decisiveness was significantly different

from zero ($t = 3.21$), whereas the analogous slope of CRN amplitudes was not significant ($t = 1.91$). To visualize these effects, Figure 6 shows model estimates for response type and decisiveness. In addition, the model showed that the effect of trial number, as well as the interaction of trial number and response type, were also significant. The slope of CRN amplitudes depending on trial number ($0.11 - 0.5 * 0.11 = 0.06$) was more attenuated than the slope of ERN amplitudes ($0.11 + 0.5 * 0.11 = 0.17$). This means that every 10 trials were accompanied by an increase of $0.06 \mu\text{V}$ in the CRN and of $0.17 \mu\text{V}$ in the ERN (trial number was divided by 10 to facilitate LME's convergence). Direct tests conducted separately for the response-error and response-correct conditions revealed that the slope of ERN amplitudes in the function of trial number was significantly different from zero ($t = 3.14$), whereas the analogous slope of CRN amplitudes was not significant ($t = 1.69$). Thus, the interaction effect between trial number and response type indicates that, as the experiment progressed, the amplitude of the ERN tended to become less pronounced.

3.4 | Exploratory analyses

3.4.1 | Onset latency of the P3 component

Since inhibition performance may manifest not only in the amplitude of the P3, but also in its latency (Kok et al., 2004;

TABLE 2 Estimates of the LME model of the P3 amplitude

Fixed effects				Random effects (by subject)
Effect	Estimate	SE	t value	SD
Intercept	13.82	1.08	12.81	7.03
Trial number	0.39	0.22	1.78	1.10
Trial type	-3.26	0.56	-5.79	2.17
Decisiveness	-1.90	0.86	-2.22	
Stop delay 100 vs. 150	-0.25	0.62	-0.40	1.52
Stop delay 100 vs. 200	2.05	0.92	2.23	3.37
Stop delay 100 vs. 250	-1.13	1.01	-1.11	1.20
Stop delay 100 vs. 300	-1.00	1.54	-0.65	4.58
Stop delay 100 vs. 350	-0.11	1.81	-0.06	3.37
Stop delay 100 vs. 400	-7.46	2.28	-3.27	2.52
Trial number:Trial type	-0.09	0.32	-0.29	
Trial type:Decisiveness	0.17	0.48	0.36	
Residual				10.96

Note. Significant effects are indicated in bold.

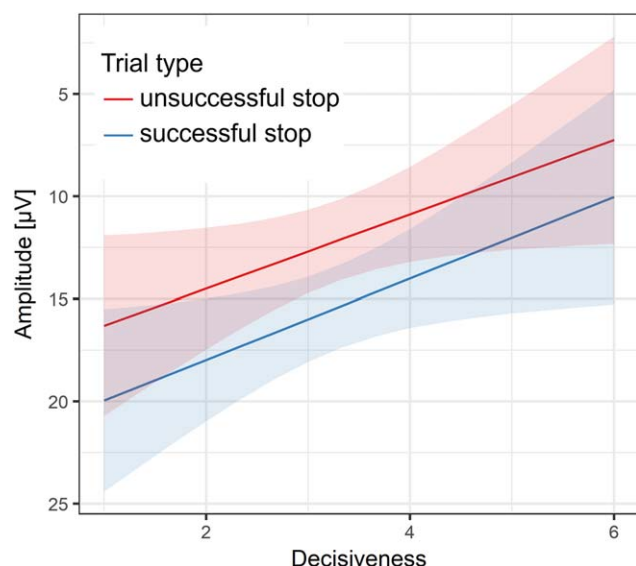


FIGURE 4 Predictions of the LME estimates of the P3 amplitude. Colored bands represent 95% confidence intervals based on the model. Values on the y axis are inverted to correspond with ERP plots

Wessel & Aron, 2015), we conducted a separate analysis measuring the onset of the P3 component.² We used the 50% peak latency method, as recommended by Luck (2014). To make the analysis of the P3 onset latency comparable to the analyses of ERP amplitudes, we established the P3 onset latency for each individual trial and used LMEs to analyze the data. We used the same fixed effects as in the analyses of stimulus-locked ERP amplitudes (N1 and P3).

The onset of the P3 component varied depending both on trial type and on decisiveness. As indicated by the significant main effects of trial type and decisiveness, trials in which participants managed to withhold their go response were characterized by P3s with a shorter onset latency. Independently, participants scoring higher on decisiveness had longer P3 onset latencies (see Table 4, Figure 7). The interaction of trial type and decisiveness, as well as all effects involving trial type, were not significant.

3.4.2 | Mediation analyses

Correlation analysis presented in the behavioral results showed that SSRT correlated positively with decisiveness. Therefore, we conducted additional analyses to determine whether SSRT mediates the relationships between decisiveness and the ERPs. For example, it is possible that it is SSRT that directly predicts P3 amplitude and decisiveness can indirectly explain the P3 amplitude because it modulates SSRT. We repeated all ERP analyses described above and added two fixed-effect terms into the analyses: SSRT and

either an interaction of SSRT and trial type (for the stop-signal-locked analyses) or an interaction of SSRT and response type (for the response-locked analysis). Because SSRT is a between-subjects factor, the by-subject random effects were left untouched (and were subjected only to the trimming of zero variance random effects).

The effects involving SSRT were significant only in the models predicting P3 amplitude and P3 latency. In the analysis of the P3 amplitude, SSRT completely replaced decisiveness in predicting the dependent variable, and the estimate of decisiveness fell from -1.9 to -0.93 , new $t = -1.09$ (see Appendix Table A1 for the full model). This indicates that SSRT fully mediates the effect of decisiveness on the P3 amplitude.

In the analysis of the P3 latency (including decisiveness, SSRT, and their interactions with trial type), both decisiveness and SSRT predicted the dependent variable with equal (and significant) reliability (see Table A2 for the full model), while the estimate of decisiveness fell from 21.39 to 15.54 ; new $t = 2.83$, relative to the model including only decisiveness. This indicates that the unique variance of SSRT and decisiveness, as well as the variance shared between these variables, all predict the P3 latency. The LME models for the mean amplitude of the N1 and ERN/CRN components, including SSRT and its interaction with trial or response type (not significant), are shown in Table A3 and A4, respectively.

4 | DISCUSSION

The aim of the present study was to examine whether response inhibition, performance monitoring, and their ERP correlates vary between individuals differing in decisiveness, as assessed by the decisiveness subscale of the NFC questionnaire (Webster & Kruglanski, 1994). Our results indicate that highly decisive people and those low in decisiveness differ in ERP measures related to stop-signal and response processing. In particular, participants with higher decisiveness scores showed reduced brain activity in the N1 and P3 time window in both successfully and unsuccessfully inhibited trials. Additionally, high decisiveness was associated with decreased negativity in the ERN time window. Regarding behavior, we observed a significant correlation between SSRT, inhibition rate, and decisiveness, indicating diminished cognitive control in highly decisive participants. Thus, our findings primarily indicate that decisiveness is negatively associated with the efficiency of both response inhibition and error monitoring.

Participants differed in several behavioral performance measures. Individuals higher in decisiveness were slower in their SSRT and less efficient in inhibiting motor responses. Long SSRT is usually implicated as a reflection of poor

²We thank the reviewers for directing our attention to analyses described in the Exploratory analyses section.

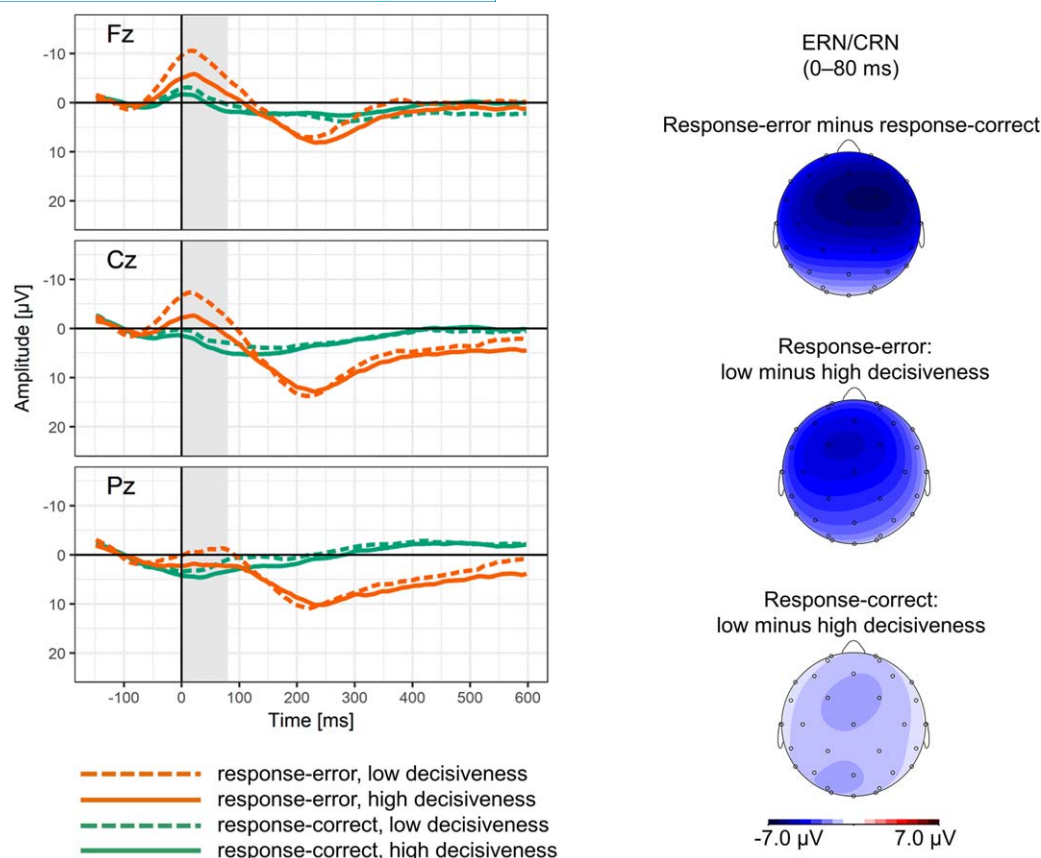


FIGURE 5 Response-locked grand-averaged waveforms from all trials at representative midline electrodes Fz, Cz, and Pz (left) with scalp potential difference maps for the ERN/CRN component (right). Two groups of participants scoring high and low on decisiveness were formed by a median split procedure only for illustrative purposes. The component-specific window examined in this study is highlighted

inhibitory control or a tendency toward impulsive responding. This assumption is supported by studies pointing to a positive correlation between SSRT and impulsivity scores (e.g., Logan et al., 1997; Stahl & Gibbons, 2007). Moreover,

no correlation was found between go RT and decisiveness. This finding suggests that, in highly decisive individuals, compromised performance was mainly limited to response inhibition, whereas their go-response performance was

TABLE 3 Estimates of the LME model of the ERN/CRN amplitude

Fixed effects				Random effects (by subject)
Effect	Estimate	SE	<i>t</i> value	SD
Intercept	-2.17	0.56	-3.87	4.01
Trial number	0.11	0.03	3.23	0.14
Response type	-6.24	0.54	-11.60	3.36
Decisiveness	1.34	0.49	2.76	
Trial number: Response type	0.11	0.05	2.10	0.05
Response type: Decisiveness	0.98	0.47	2.08	
Residual				10.37

Note. Significant effects are indicated in bold.

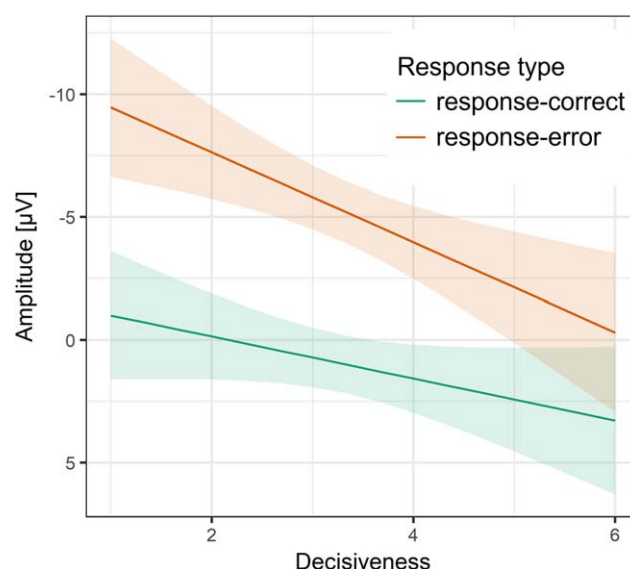


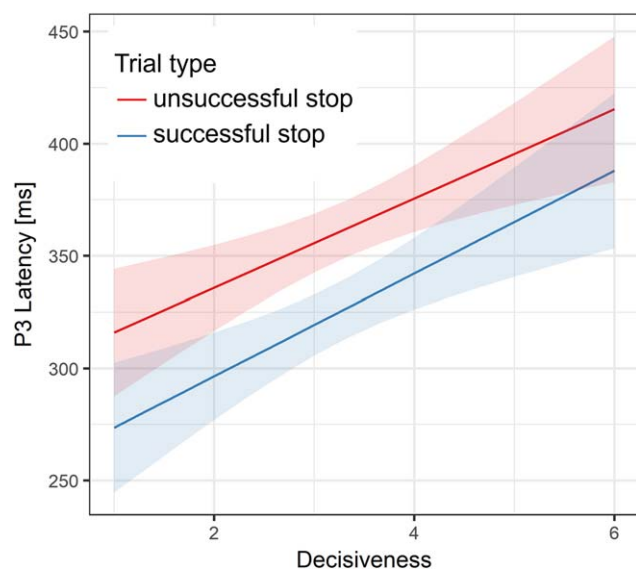
FIGURE 6 Predictions of the LME estimates of the ERN/CRN amplitude. Colored bands represent 95% confidence intervals based on the model. Values on the y axis are inverted to correspond with ERP plots

TABLE 4 Estimates of the LME model of the P3 onset latency

Fixed effects				Random effects (by subject)
Effect	Estimate	SE	<i>t</i> value	SD
Intercept	343.22	7.24	47.41	43.55
Trial number	1.25	1.67	0.75	5.94
Trial type	35.57	4.46	7.98	7.68
Decisiveness	21.39	5.44	3.93	
Stop delay 100 vs. 150	4.16	5.29	0.79	
Stop delay 100 vs. 200	1.99	7.17	0.28	13.69
Stop delay 100 vs. 250	13.88	8.60	1.61	
Stop delay 100 vs. 300	−4.19	11.55	−0.36	20.99
Stop delay 100 vs. 350	−19.29	14.31	−1.35	
Stop delay 100 vs. 400	10.09	18.43	0.55	
Trial number:Trial type	1.05	2.93	0.36	2.07
Trial type:Decisiveness	−2.99	3.80	−0.79	
Residual				100.56

Note. Significant effects are indicated in bold.

similar (at least on a behavioral level) to that observed in participants lower in decisiveness. Disproportionally long SSRTs without significant differences in go RTs have frequently been observed in adults with low impulse control (Lijffijt et al., 2005).

**FIGURE 7** Predictions of the LME estimates of the P3 onset latency. Colored bands represent 95% confidence intervals based on the model

4.1 | Stop-signal processing

The N1 component was more enhanced in successful than in failed stop trials; this points to an increased activation of sensory areas and aligns with the findings of previous SST studies (Bekker, Kenemans et al., 2005; Dimoska & Johnstone, 2008; Hughes, Fulham, Johnston, & Michie, 2012; Lansbergen et al., 2007; Senderecka, 2016, 2018). In addition, stop signals evoked less pronounced N1 in individuals scoring higher on decisiveness, independently of the stop-signal trial type. This suggests that high decisiveness was associated with a less efficient shift of attention toward a stop signal on its presentation, which was probably partly determinative of the subsequent response inhibition (Bekker, Kenemans et al., 2005). It is worth noting that task-set switching deficits have been previously demonstrated in impulsive individuals (Bekker, Overtoom et al. 2005; Cepeda, Cepeda, & Kramer, 2000). A potential explanation for our finding is that, in the case of highly decisive individuals, go-stimuli processing interfered with stop-signal detection to a greater extent than in those low in decisiveness. This could lead to less effective attentional switching driven in a bottom-up manner from the ongoing motor planning to the rarely presented inhibition cues (Kenemans, 2015). This explanation is supported by the fact that participants were equally fast in go RTs, which suggests that individual differences in behavioral performance were confined to SSRTs. However, since the stop signal was

never presented alone, we cannot exclude the possibility that highly decisive individuals might also show reduced auditory N1 in settings that do not require attentional switching between stimuli presented in rapid succession. Further research is surely needed to attain a thorough understanding of the associations between initial stages of sensory analysis and decisiveness.

The P3 amplitude was larger for the successful than for the failed stop trials; this is a common result in SST studies (e.g., De Jong et al., 1990; Dimoska, Johnstone, & Barry, 2006; Greenhouse & Wessel, 2013; Hughes et al., 2012; Overtom et al., 2002; Senderecka, 2016, 2018). The mixed-effects analysis revealed that, in line with what we observed for the N1 component, stop signals evoked less pronounced P3 in highly decisive individuals in both successfully and unsuccessfully inhibited trials. This finding suggests that highly decisive people may have reduced efficiency of the neural mechanisms responsible for the suppression and slowing of motor behavior (Bekker, Kenemans et al., 2005; De Jong et al., 1990), similar to individuals with low impulse control (e.g., Bekker, Overtom et al. 2005; Liotti et al., 2005; Overtom et al., 2002; Senderecka et al., 2012). However, it should be noted that, when SSRT was included in the model, the direct effect of decisiveness on the P3 amplitude turned out to be not significant, indicating that participants' SSRT fully mediated the relationship between decisiveness and P3 amplitude. In other words, decisiveness indirectly influenced the P3 amplitude by modulating SSRT. Thus, our results suggest that high decisiveness adversely impacts the recruitment of brain circuits responsible for motor suppression (as indexed by P3 amplitude), but this impact is mediated by increased latency of reactions to the stop signal.

Additionally, we observed that the P3 onset latency was shorter for successful compared to failed stop trials; this is in line with the findings of previous research (Wessel & Aron, 2015, see also Kok et al., 2004, for the P3 peak latency analyses) and supports the notion that timely activation of the stop process is essential for successful inhibition. This effect is entirely consistent with the horse race model, thus implying that the probability of inhibition depends on a race between independently operating go and stop processes (Logan & Cowan, 1984). Importantly, we found a positive association between decisiveness and P3 onset latency, suggesting that the P3 onset was delayed in highly decisive people. This implies that individuals scoring higher on decisiveness, compared to people scoring lower on this measure, are slower to engage the stopping process. Mediation analysis revealed that both decisiveness and SSRT significantly predicted the P3 onset latency. Thus, the late P3 onset in highly decisive participants is independently related to both high decisiveness and prolonged duration of the stopping process.

4.2 | Response processing

In line with previous research (Falkenstein et al., 1991; Nieuwenhuis, Ridderinkhof, Blom, Band, & Kok, 2001), the early negativity observed in the ERPs time-locked to responses was larger for the erroneous than for the correct responses. An interesting finding of our study is that high decisiveness was associated with decreased ERN amplitude (as indicated by the interaction effect between response type and decisiveness). As has been suggested by Stahl and Gibbons (2007), the most relevant theoretical approach for the ERN in the SST context is the response-conflict monitoring theory (Botvinick et al., 2001; Carter & van Veen, 2007; Yeung et al., 2004). According to this account, the ERN reflects a mechanism that monitors ongoing response conflicts that occur when two or more response options are simultaneously activated. The amount of conflict between a competing fast erroneous response and a subsequent, slower corrective response is greatest immediately after error commission (i.e., when the activations of both the correct and incorrect response overlap in time). These conflict detection and action monitoring processes are presumably executed by the anterior cingulate cortex (Ridderinkhof, Ullsperger, Crone, & Nieuwenhuis, 2004). Since more negative ERN amplitudes are associated with increased levels of response conflict (Carter & van Veen, 2007; Yeung et al., 2004), our findings may indicate that people scoring higher on decisiveness have a smaller degree of postresponse conflict. In turn, this may suggest that highly decisive individuals have weakened representation of stop-signal-appropriate responses or generally less efficient monitoring of responses when the requirements of the task are context dependent.

Since several accounts have been proposed regarding the functional significance of the ERN, the present data should also be discussed within other frameworks.³ An important approach to the ERN is provided by the mismatch theory (Coles et al., 2001; Falkenstein et al., 1991; Gehring et al., 1993), which assumes that the ERN represents an error signal that occurs when the actual response does not match the correct response. The larger the mismatch between the actually performed action and the desired state, the more pronounced the ERN. Considering this theory, our results may suggest that the mismatch detection process is less effective in people high in decisiveness.

Additionally, it has been proposed that the ERN is related to the subjective significance and aversiveness of an error (Hajcak et al., 2005; Hajcak & Foti, 2008). Following this line of interpretation, the association between smaller ERNs and higher decisiveness might indicate that errors were less

³Since the reinforcement learning theory (Holroyd & Coles, 2002) focuses on the processes that follow rather than accompany an error, we do not discuss this account.

significant and less distressing for highly decisive participants. Previous studies characterized highly decisive people as less likely to interpret ambiguous situations as threatening (Rassin & Muris, 2005) and pointed to the negative association between neuroticism, anxiety, and decisiveness (Germeijs et al., 2006; Germeijs & Verschuere, 2011). These findings become particularly interesting when compared with the results of several ERP studies indicating an association between increased error monitoring and negative emotionality. For example, a reliably enhanced ERN amplitude has been observed in patients with anxiety disorders (Aarts & Pourtois, 2010; Hajcak, McDonald, & Simons, 2003), among nonclinical individuals with high levels of neuroticism or negative affect (Hajcak, McDonald, & Simons, 2004; Luu, Collins, & Tucker, 2000; Pailing & Segalowitz, 2004b), with increased punishment sensitivity (Boksem, Tops, Koster, & De Cremer, 2008; Boksem, Tops, Wester, Meijman, & Lorist, 2006), and experiencing of cognitive distress in the face of uncertainty (Jackson, Nelson, & Hajcak, 2016). Since in our study higher decisiveness was associated with a decreased ERN, our results indirectly support conclusions from previous research, which highlighted an inverse association between decisiveness and negative affectivity (Germeijs et al., 2006; Germeijs & Verschuere, 2011; Rassin & Muris, 2005).

It should be noted that these different accounts of the ERN, including theories of conflict monitoring, mismatch detection, and emotional/motivational responses to errors, are not mutually exclusive. Rather, they emphasize different aspects of the cognitive-affective system responsible for goal-directed behavior. Thus, we do not purport to adjudicate between these theories with our current data. Instead, we accept that there are several plausible interpretations to explain reduced ERN amplitudes in highly decisive individuals, such as a smaller degree of postresponse conflict, compromised error detection, or less subjective significance of an error.

It is worth noting that a largely overlapping pattern of results that indicates reduced ERN amplitudes was found in individuals with a tendency toward impulsive responding (Pailing, Segalowitz, Dywan, & Davies, 2002; Ruchow et al., 2005; Stahl & Gibbons, 2007) or with high self-reported impulsivity (Martin & Potts, 2009; Potts, George, Martin, & Barratt, 2006; Taylor, Visser, Fueggle, Bellgrove, & Fox, 2018). Thus, taken together with the previous findings, the present data support the assumption that both decisiveness and impulsivity may represent a maladaptive extreme of decision making that is linked to low response control and reduced error monitoring.

Additionally, our analyses revealed that the ERN became more flattened as the experiment progressed. There is broad agreement that response-related ERPs, particularly the ERN, are sensitive to the degree of involvement during a task and

can reflect state variations in engagement (Boksem, Meijman, & Lorist, 2006; Luu et al., 2000; Tops & Boksem, 2010; Tops, Boksem, Wester, Lorist, & Meijman, 2006). Since intrinsic engagement is highest at the start of a performance, ERN amplitudes are usually largest at the beginning and decrease over the course of a task. This is especially true for fatiguing and demanding paradigms (such as SST) or monotonous tasks that require continuous performance for a long period of time. The lack of a significant relationship between decisiveness and the ERN amplitude reduction that proceeded from the start to the end of our task suggests that initial high engagement and subsequent disengagement from action monitoring was homogeneous across the whole sample. Therefore, it is reasonable to conclude that the self-reported level of decisiveness is not sufficient to predict the deterioration of performance monitoring during the task.

4.3 | Conclusions and future directions

To our knowledge, our study is the first to provide evidence of the negative association between decisiveness and the efficiency of both response inhibition and error monitoring. Our results suggest that decisiveness, similar to impulsivity, is a trait reflecting behavioral control. Usually, highly impulsive individuals act without much deliberation, which often results in error-prone responding (Dickman, 1990). Such a response strategy was observed in our study in highly decisive individuals, as evidenced by prolonged SSRT and decreased inhibition rate accompanied by reduced amplitudes of N1, P3, and ERN. However, two limitations of the present study should be mentioned here. First, our results are not sufficient to determine whether there is a direct link between decisiveness and diminished cognitive control. The association between these two factors may be driven by a third variable such as subclinical impulsivity symptoms, which were not directly measured in our study. Second, decisiveness may manifest itself in distinct ways (Barkley-Levenson & Fox, 2016; Janis & Mann, 1977), and various subtypes of decisiveness may exhibit different relationships with neural and behavioral indices of response inhibition and performance monitoring. Thus, it would be worthwhile to replicate the present results using a greater variety of decisiveness measures.

Decisiveness is an important aspect of adaptive decision making that can, depending on the situation, lead to optimal or suboptimal decision outcomes. Although clearly more research is needed to attain a better understanding of the associations between decisiveness and cognitive control, our study provides evidence that the behavioral tendencies of highly decisive individuals may stem from low response inhibition and weak performance monitoring, both of which

are accompanied by relatively low activity of the brain areas subserving these functions.

ACKNOWLEDGMENTS

The research was funded by an Iuventus Plus grant (0486/IP3/2011/71) from the Polish Ministry of Science and Higher Education (data collection) and an Opus 10 grant (2015/19/B/HS6/00341) from the National Science Centre of Poland awarded to Magdalena Senderecka. During the work on the project, Małgorzata Kossowska was supported by a Maestro grant (DEC 2011/02/A/HS6/00155) from the National Science Centre of Poland. The funding sources had no impact on any part of the present study. The authors gratefully acknowledge the help of Ewelina Frackiewicz, Patrycja Kałamała, and Tomasz Ligeza with data recording. The authors are also grateful to the participants who volunteered to take part in the study and to Michael Timberlake who proofread the manuscript. EEG recording was possible thanks to the cooperation of LangUsta, Psychology of Language and Bilingualism Lab, Institute of Psychology, Jagiellonian University, Kraków.

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How to cite this article: Senderecka M, Szewczyk J, Wichary S, Kossowska M. Individual differences in decisiveness: ERP correlates of response inhibition and error monitoring. *Psychophysiology*. 2018;55:e13198. <https://doi.org/10.1111/psyp.13198>

APPENDIX : LME MODELS INCLUDING SSRT AND ITS INTERACTIONS—MEDIATION ANALYSES

TABLE A1 Estimates of the LME model of the P3 amplitude—Mediation analysis

Fixed effects			
Effect	Estimate	SE	t value
Intercept	13.65	1.01	13.46
Trial number	0.40	0.22	1.81
Trial type	−3.22	0.56	−5.77
Decisiveness	−0.93	0.86	−1.09
SSRT	−0.06	0.02	−2.93
Stop delay 100 vs. 150	−0.38	0.62	−0.61
Stop delay 100 vs. 200	1.82	0.92	1.97
Stop delay 100 vs. 250	−1.42	1.01	−1.40
Stop delay 100 vs. 300	−1.42	1.55	−0.91
Stop delay 100 vs. 350	−0.52	1.82	−0.29
Stop delay 100 vs. 400	−8.01	2.23	−3.59
Trial number:Trial type	−0.09	0.32	−0.26
Trial type:Decisiveness	0.44	0.50	0.86
Trial type:SSRT	−0.02	0.01	−1.35

Note. Significant effects are indicated in bold.

TABLE A2 Estimates of the LME model of the P3 onset latency—Mediation analysis

Fixed effects			
Effect	Estimate	SE	t value
Intercept	344.46	6.91	49.87
Trial number	1.16	1.67	0.69
Trial type	34.86	4.49	7.76
Decisiveness	15.54	5.49	2.83
SSRT	0.36	0.13	2.82
Stop delay 100 vs. 150	5.29	5.30	1.00
Stop delay 100 vs. 200	3.85	7.19	0.54
Stop delay 100 vs. 250	16.38	8.62	1.90
Stop delay 100 vs. 300	−1.11	11.58	−0.10
Stop delay 100 vs. 350	−16.23	14.33	−1.13
Stop delay 100 vs. 400	13.49	18.48	0.73
Trial number:Trial type	1.08	2.94	0.37
Trial type:Decisiveness	−3.86	4.03	−0.96
Trial type:SSRT	0.04	0.10	0.39

Note. Significant effects are indicated in bold.

TABLE A3 Estimates of the LME model of the N1 amplitude—Mediation analysis

Fixed effects			
Effect	Estimate	SE	t value
Intercept	−7.64	0.77	−9.87
Trial number	−0.27	0.17	−1.61
Trial type	4.30	0.51	8.38
Decisiveness	1.33	0.62	2.16
SSRT	0.00	0.01	−0.14
Stop delay 100 vs. 150	3.42	0.59	5.76
Stop delay 100 vs. 200	2.63	0.84	3.15
Stop delay 100 vs. 250	−1.09	1.09	−1.00
Stop delay 100 vs. 300	−2.95	1.13	−2.61
Stop delay 100 vs. 350	−0.81	2.43	−0.33
Stop delay 100 vs. 400	−5.43	1.90	−2.85
Trial number:Trial type	−0.05	0.30	−0.18
Trial type:Decisiveness	−0.58	0.47	−1.24
Trial type:SSRT	0.00	0.01	−0.08

Note. Significant effects are indicated in bold.

TABLE A4 Estimates of the LME model of the ERN/CRN amplitude—Mediation analysis

Fixed effects			
Effect	Estimate	SE	t value
Intercept	−2.16	0.56	−3.89
Trial number	0.10	0.03	3.22
Response type	−6.25	0.54	−11.58
Decisiveness	1.06	0.52	2.02
SSRT	0.02	0.01	1.41
Trial number:Response type	0.11	0.05	2.09
Response type:Decisiveness	0.83	0.51	1.61
Response type:SSRT	0.01	0.01	0.72

Note. Significant effects are indicated in bold.